

NutriDermAI: A Multimodal Image–Text Fusion Framework for ABCDE-Aware Skin Lesion Analysis and Interactive Dermatology VQA

Akhil Kulal

Department of Computer Science
Institute / University Name
Email: akhil.kulal@example.com

Prof. Manasi Kulkarni

Department of Computer Science
Institute / University Name
Email: manasi.kulkarni@example.com

Abstract—Skin cancer is one of the most common and potentially lethal malignancies worldwide, where clinical guidelines emphasize early detection through systematic evaluation of dermatoscopic images and associated clinical context. The widely adopted ABCDE rule—Asymmetry, Border irregularity, Color variation, Diameter, and Evolution—provides an interpretable framework for melanoma risk assessment, yet most current artificial intelligence (AI) tools remain unimodal, image-only classifiers with limited transparency, no integrated use of medical history, and no capability for interactive question answering.

This paper proposes *NutriDermAI*, a multimodal dermatology assistant that unifies (i) skin lesion segmentation, (ii) lesion region-of-interest (ROI) extraction, (iii) hierarchical lesion classification with image-based ABCD computation, (iv) medical text understanding for deriving “E” and clinical context, (v) an Image–Text Fusion (IT-Fusion) layer for calibrated ABCDE risk scoring, and (vi) a Visual Question Answering (VQA) module for interactive, patient-centric and clinician-centric dialogue. The system is implemented as a nine-stage pipeline, covering dataset preprocessing, model training, multimodal fusion, conversational reasoning, and deployment. We describe the architecture, stage-wise design, target datasets, planned models, and evaluation protocol, positioning *NutriDermAI* against recent advances in dermatology foundation models, multimodal medical AI, and medical VQA. The work is framed as a proposal for a deployable, explainable, and safety-aware dermatology support system suitable for conference-level publication and future clinical validation.

Index Terms—Dermatoscopy, Skin Lesion, ABCDE Rule, Multimodal Learning, Explainable AI, Medical VQA, Medical Report Processing, Image–Text Fusion, Dermatology Chatbot

I. INTRODUCTION

Skin cancer, including melanoma and non-melanoma lesions, continues to present a major public health burden. Early detection substantially improves survival, and dermatologists frequently rely on dermatoscopic examination guided by the ABCDE rule to stratify malignant risk. Classical ABCD/ABCDE-based expert systems and rule-driven scoring approaches have demonstrated that structured feature assessment (e.g., asymmetry, border irregularity, color variegation, diameter, evolution) can approximate dermatologists’ diagnostic reasoning [1]–[5].

Concurrently, deep learning has achieved high performance on dermatoscopic segmentation and classification tasks using

U-Net variants, hybrid CNN–Transformer designs, and large-scale dermatology datasets such as HAM10000, ISIC 2016–2024, BCN20000, and MiLK-10K [6]–[17]. More recently, dermatology foundation models and multimodal vision–language models (VLMs) such as PanDerm, MM-Skin, and DermIM have shown how large-scale, expert-curated multimodal supervision improves zero-shot recognition and text grounding in dermatology [18]–[22].

In parallel, medical natural language processing (NLP) and medical VQA have evolved rapidly, leveraging clinical corpora like MIMIC-III/IV and specialized architectures such as BioBERT, medical multimodal large language models (MM-LLMs), and reinforcement learning (RL)-trained VQA systems [23]–[33]. Dermatology-specific VQA benchmarks and chat-based assistants are emerging, yet they typically lack end-to-end ABCDE computation and structured fusion of imaging with clinical text [34]–[44].

Motivation. Existing dermatology AI solutions can (i) segment lesions, (ii) classify lesions into benign/malignant or multi-class taxonomies, or (iii) support generic multimodal question answering. However, there is still a gap for a unified dermatology assistant that:

- Computes explicit ABCDE values in a manner aligned with clinical ABCDE literature and scoring rules;
- Integrates dermatoscopic images with patient symptoms, history, and laboratory or biopsy reports;
- Produces calibrated risk scores and structured JSON explanations usable by downstream applications;
- Supports safe, context-aware VQA with explicit rationales grounded in ABCDE features and clinical evidence.

Contribution. This paper proposes *NutriDermAI*, a nine-stage multimodal ABCDE-aware pipeline that:

- 1) Preprocesses image and text datasets for training and deployment (Stage 1).
- 2) Performs skin lesion segmentation (SLS) and lesion region-of-interest (ROI) extraction (Stages 2–3).
- 3) Learns hierarchical lesion classification and extracts image-based ABCD metrics (Stage 4).

- 4) Processes medical history and reports to derive evolution (E) and clinical context (Stage 5).
- 5) Fuses all evidence in an IT-Fusion layer to compute final ABCDE, risk scores, and explanations (Stage 6).
- 6) Uses a dermatology-aware VQA module for interactive dialogue with patients and clinicians (Stage 7).
- 7) Integrates frontend, backend, and deployment components with appropriate testing (Stages 8–9).

The paper is structured as a proposal-style research article suitable for conferences, outlining problem definition, related work, architecture, datasets, implementation plan, evaluation methodology, and ethical considerations.

II. BACKGROUND AND RELATED WORK

A. ABCDE Rule and Classical Dermatology Decision Support

The ABCD/ABCDE rule offers a structured approach to melanoma risk assessment by quantifying asymmetry, border irregularity, color variation, diameter, and evolution over time. Classical expert systems and analytic pipelines compute scores from dermoscopic images and combine them into total dermoscopic scores (TDS) [1]–[3]. Recent work has shown that generative models and AI systems that explicitly incorporate ABCDE analysis can match or approach dermatologist-level decision-making and provide interpretable outputs [4], [5]. Hybrid CNN+ABCD models also demonstrate that explicit ABCD descriptors can boost classification performance when coupled with deep features [45].

B. Skin Lesion Segmentation

Lesion segmentation is a prerequisite for many downstream tasks, including ABCDE feature computation and ROI cropping. U-Net and its numerous variants remain dominant in medical image segmentation [6], [46]. For skin lesions, dermatology-specialized architectures such as ARC-UNet, ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and dual CNN–ViT hybrids have improved boundary precision, small-lesion sensitivity, and robustness by leveraging residual connections, deformable attention, multiscale inputs, and metadata [7]–[13], [47]. Multiscale Swin Transformer-based models further improve segmentation across diverse medical imaging domains [48], [49].

C. Lesion Classification and Dermatology Foundation Models

Large dermoscopic datasets—including ISIC 2016–2024 challenges, HAM10000, BCN20000, and MiLK-10K—have enabled high-performing lesion classifiers and foundation models [14]–[17]. Efficient transformer-based backbones such as EfficientFormerV2 provide competitive accuracy with reduced memory footprint [50], making them suitable for multi-stage pipelines. Recent dermatology foundation models like PanDerm, MM-Skin, and DermIM-scale vision–language datasets align dermoscopic images with rich textual descriptions and hierarchical ontologies [18]–[21]. These works motivate the use of shared multimodal encoders for segmentation, classification, and fusion.

D. Clinical Text Understanding and Medical NLP

Clinical narratives, pathology reports, and lab summaries contain crucial contextual information—including symptom evolution and risk factors. BioBERT and related biomedical-language encoders have improved named entity recognition and classification over biomedical literature and clinical notes [23]. Large ICU corpora such as MIMIC-III and MIMIC-IV Note provide rich free-text and structured data for training and evaluation [24], [25]. Systematic reviews of clinical text data highlight both opportunities and pitfalls for machine learning on noisy, heterogeneous EHR text [26]. These insights inform our MedProc stage, where symptom descriptions, timelines, and lab values are normalized into structured features.

E. Medical Multimodal LLMs and VQA

Medical VQA research has produced datasets such as VQA-RAD, PathVQA, TM-PathVQA, HEALMQA, and dermatology-specific DermaVQA, which cover diverse modalities and question types [34]–[37], [51]. Model architectures range from BLIP-style multimodal encoders (BioMedBLIP) to specialized VQA fusion layers with semantic attention [27], [28]. Dynamic prompt adaptation and RL-based optimization have been proposed to reduce hallucinations and improve reasoning in medical VQA [29]–[32], [52].

Dermatology-focused works demonstrate joint learning of segmentation and VQA, CLIPSeg-based multimodal fusion, and knowledge-enhanced ensembles [38], [39], [53]–[55]. NutriDermAI is aligned with these directions but adds explicit ABCDE reasoning, rule-based cross-checks, and a unified JSON explanation layer.

F. Dermatology Chatbots and Conversational Diagnostic AI

Several works have explored dermatology chatbots combining CNN-based lesion analysis with rule-based triage logic [40], [41]. Broader diagnostic LLMs like AMIE evaluate conversational diagnostic quality across multiple specialties and emphasize safety, uncertainty communication, and escalation strategies [43]. Reviews of ChatGPT-style models in dermatology and multimodal LLMs in medicine further highlight the need for guardrails, bias mitigation, and rigorous evaluation [33], [42]. Studies on conversational agents in dermatology show improved patient engagement but also underline responsibility and ethical concerns [44]. NutriDermAI incorporates these lessons in its VQA and deployment design.

G. Gap and Design Requirements

From this literature, key gaps emerge:

- ABCDE-aware systems that compute all five criteria in a clinically consistent, data-driven manner remain rare.
- Integration of dermoscopic images with detailed medical history and laboratory reports is limited, especially in dermatology.
- Medical VQA systems often lack explicit links between answers and structured evidence such as ABCDE scores or rule-based checks.

- End-to-end dermatology assistants spanning segmentation, classification, multimodal fusion, VQA, and production deployment are not widely documented.

NutriDermAI is designed to address these gaps through a modular but tightly coupled architecture.

III. SYSTEM OVERVIEW

NutriDermAI is implemented as a nine-stage pipeline (Fig. 1). The diagram shows two logical layers:

- **Frontend container** (top of Fig. 1): the patient interacts with a chatbot-style UI, uploads images and reports through an upload manager, and sends questions via a REST API client. The result display component renders the final answer and evidence.
- **Backend container** (bottom of Fig. 1): a FastAPI service coordinates the core stages—segmentation (SLS), ROI cropping (SLRC), hierarchical classification with ABCD (HC), MedProc, IT-Fusion, and VQA. Intermediate artifacts such as masks, ROIs, ABCD values, symptoms, and risk JSON are persisted to a database and reused by downstream modules.

The arrows in Fig. 1 highlight the data flow: the patient’s question and uploaded data first reach the backend via an API call; the backend processes the image through SLS and SLRC, then HC, processes the text through MedProc, fuses everything in IT-Fusion, and finally VQA generates an answer. The answer, together with the evidence JSON, is returned to the frontend for display.

All stages exchange structured artifacts (e.g., binary masks, ROIs, ABCD scores, symptom vectors, unified JSON) to support transparency, modular retraining, and debugging.

IV. STAGE-WISE METHODOLOGY

A. Stage 1: Dataset and Input Preprocessing

Stage 1 prepares both training datasets and live user inputs for downstream stages. The diagram in Fig. 2 has a deliberately simple structure: a single input block at the top, an internal processing block, and an output block.

The **input block** lists the three supported raw modalities: dermatoscopic images, PDF reports, and plain text (e.g., typed history). These can be noisy (different resolutions, presence of artefacts, OCR errors).

The **task block** in the center of Fig. 2 summarizes the processing performed:

- *Image resize/normalize*: images are resized to a fixed resolution (e.g., 512×512 for SLS, 224×224 for HC) and normalized channel-wise. Optional hair removal and contrast enhancement can be added here.
- *OCR for PDFs*: scanned reports go through OCR to extract machine-readable text; metadata such as page numbering and section headers can be preserved.
- *Text cleanup*: OCR output and typed text are de-noised (removing headers, page numbers, and boilerplate), lower-cased, tokenized, and truncated to a safe length (e.g., ≤ 512 tokens for transformer-based models).

The **output block** explicitly shows that Stage 1 provides two clean streams back to the pipeline: (i) clean text suitable for MedProc, and (ii) preprocessed images suitable for SLS, SLRC, and HC. At MTech-project level, these blocks correspond to separate Python modules (e.g., `image_preproc.py` and `text_preproc.py`) invoked by the backend.

B. Stage 2: Skin Lesion Segmentation (SLS)

Given a full skin lesion image, Stage 2 produces a pixel-wise lesion mask. The SLS diagram in Fig. 3 is a vertical flow showing how a raw skin image is converted into a segmentation mask through ARC-UNet.

At the top, the **input block** specifies a single skin image (size ≤ 5 MB) coming from Stage 1. The next block, **Preproc: Resize 512×512 , Normalize, Hair removal**, concretely states the image operations applied before feeding the CNN. The **ARCUNet Segmentation CNN** block represents the main model: a U-Net variant with attention and residual connections [7]. The bottom block shows the **Output: Lesion Mask (512×512)**, which is a binary mask aligned with the original (resized) image.

In implementation terms, this stage is a standard PyTorch model trained with Dice and cross-entropy loss on ISIC segmentation labels; the diagram clarifies that the only external dependency is a preprocessed RGB input and the only output is the mask tensor, plus derived geometric features used later for ABCD computation.

C. Stage 3: Skin Lesion ROI Cropping (SLRC)

Stage 3 uses the SLS mask to derive a tight ROI around the lesion. The SLRC diagram in Fig. 4 makes explicit that this stage combines the mask from SLS and the original image.

The two **input blocks** at the top correspond to:

- *Lesion Mask (From SLS)*: the binary mask output from Stage 2.
- *Original Image*: the preprocessed dermatoscopic image aligned with the mask.

The central **Bounding Box** block describes the algorithmic core: using OpenCV connected components analysis on the mask to identify the largest lesion region, then computing its minimal bounding rectangle. The following block, **Resize 224×224** , emphasizes that this cropped patch is rescaled to a fixed input size for EfficientFormerV2.

The bottom **Output block** explicitly labels the deliverable ROI image (224×224). In code, this stage is a deterministic image-processing step that produces both the resized ROI and metadata (bounding box coordinates, scaling factors), mirroring exactly the diagram.

D. Stage 4: Hierarchical Classification and Raw ABCD Extraction (HC)

Stage 4 is more complex: it simultaneously performs classification and ABCD feature extraction. Fig. 5 visualizes this as a large rounded container with multiple internal pipelines running in parallel.

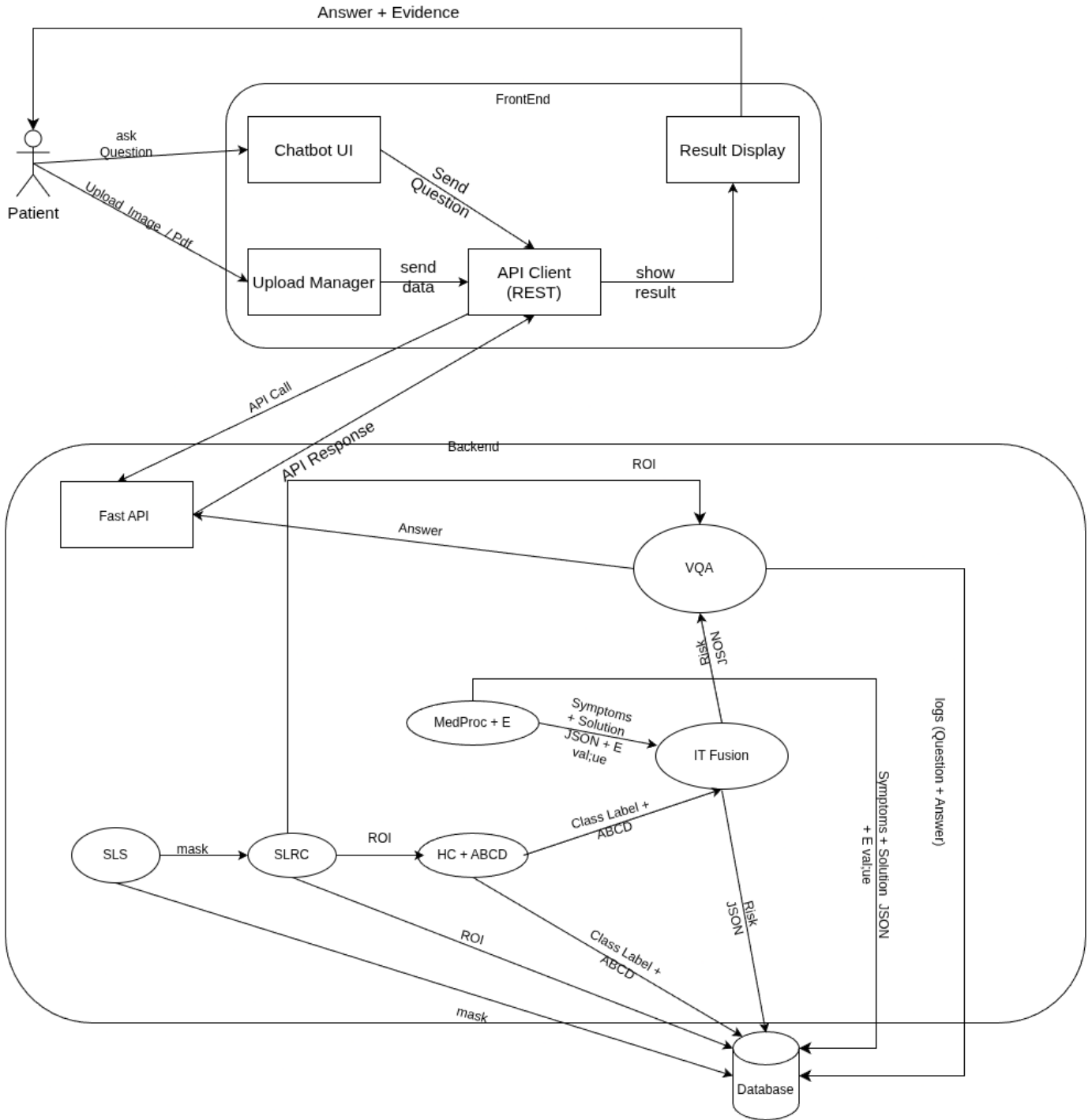


Fig. 1: High-level NutriDermAI architecture: interaction between patient-facing frontend and multi-stage backend (SLS, SLRC, HC+ABCD, MedProc+E, IT-Fusion, VQA) with database logging.

On the left of Fig. 5, a vertical path shows the **classification branch**:

- The *Input: ROI Image (SLRC)* block receives the 224×224 patch from Stage 3.
- The *Preproc: Normalize, Augment (Train only)* block summarizes standard data augmentation (flips, rotations) and

normalization.

- The *EfficientFormerV2-S Vision Transformer* block denotes the backbone used for hierarchical classification [50].
- The *Class Label* block at the bottom produces benign/malignant and multi-class predictions.

The four vertical columns to the right correspond to the

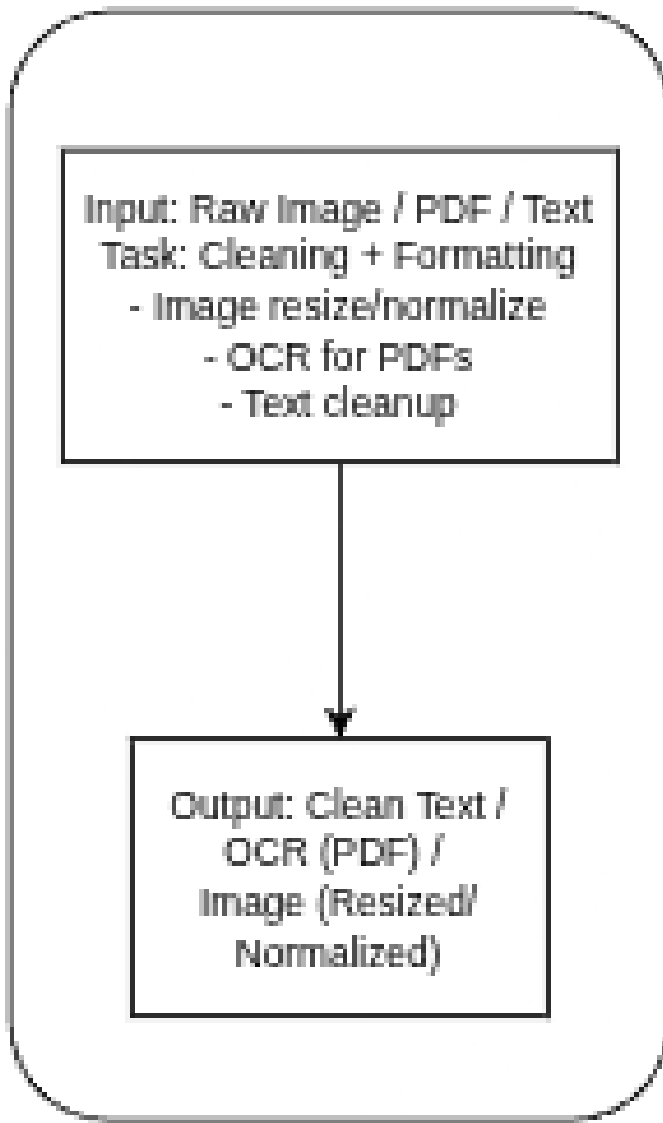


Fig. 2: Stage 1 (Preprocessing): cleaning and formatting heterogeneous image, PDF, and text inputs into standardized clean image and text outputs.

ABCD sub-pipelines:

- **A: Asymmetry** column takes the lesion mask as input and applies symmetry analysis by mirroring the mask across two axes and computing area overlap. The output is a scalar A-value.
- **B: Border Irregularity** column also uses the mask to derive contour complexity measures such as compactness and fractal dimension, producing a B-value.
- **C: Color Variance** column uses the ROI image to perform color clustering (e.g., *k*-means in LAB space) and quantifies how varied the colors are, yielding a C-value.
- **D: Diameter** column computes the maximum distance between any two pixels in the lesion mask and scales it (when calibration is available) to an approximate diameter in millimeters, giving a D-value.

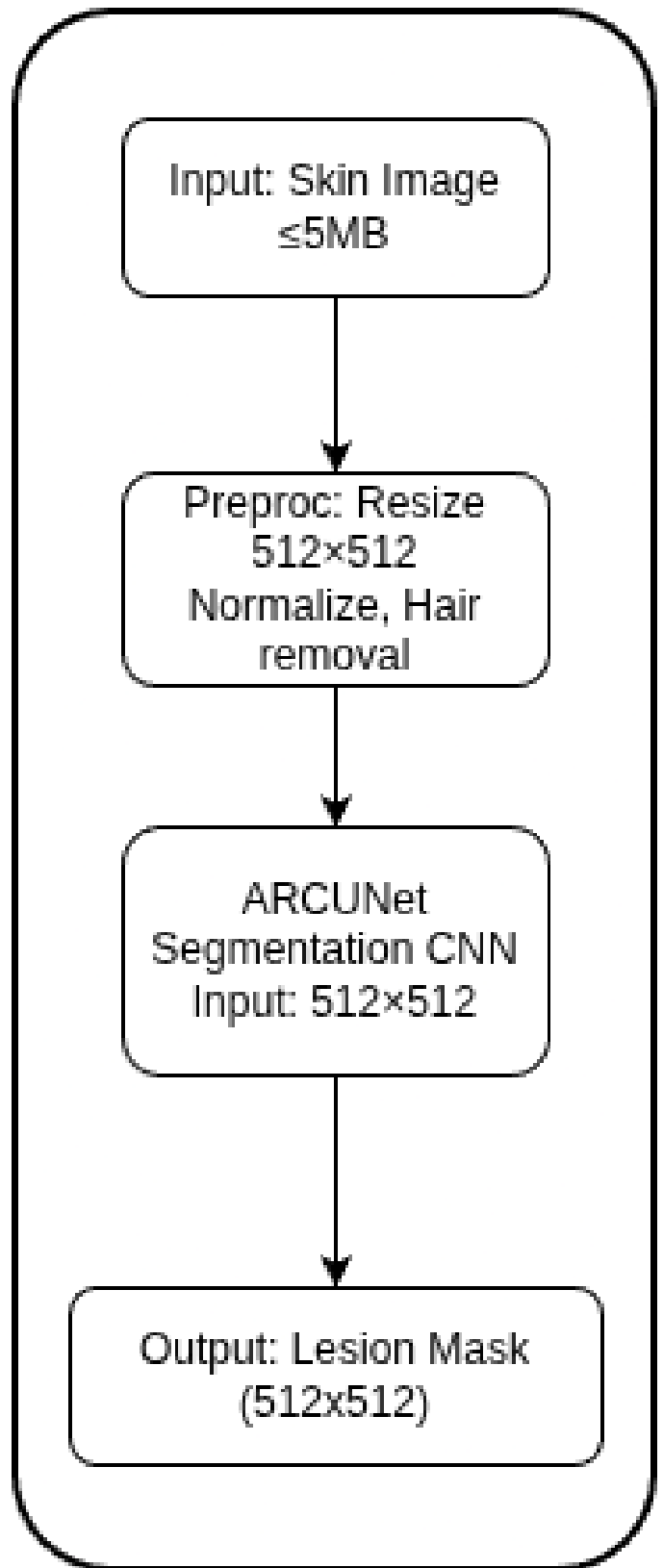


Fig. 3: Stage 2 (SLS): ARC-UNet based segmentation from preprocessed dermatoscopic images to 512×512 lesion masks.

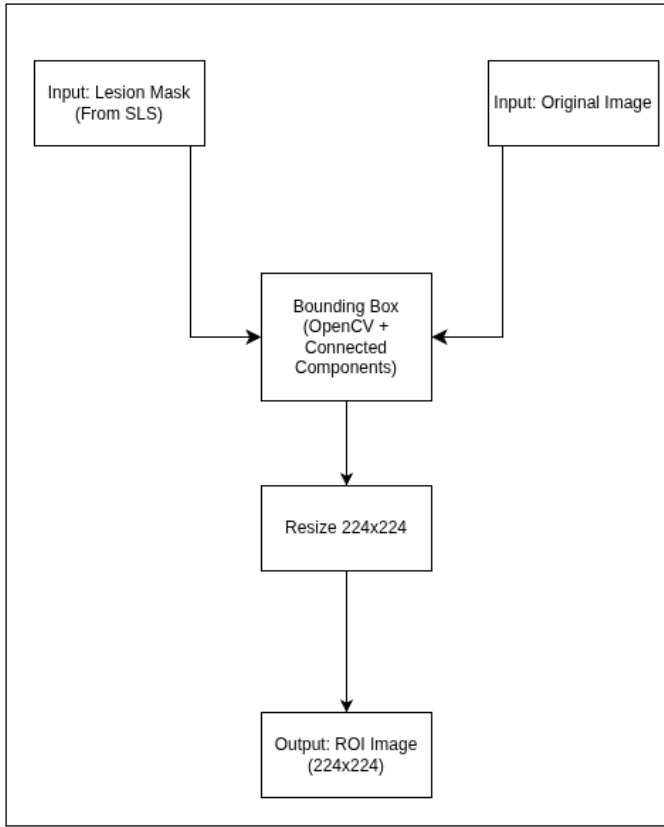


Fig. 4: Stage 3 (SLRC): mask-guided bounding-box extraction and resizing to produce a standardized ROI image.

All four values feed into an **ABCD Aggregator** block, which combines the A–D features and is then linked back to the classification path. The final **Output block** at the bottom of Fig. 5 therefore returns both the class label and explicit ABCD values. For an MTech project implementation, each column in the figure corresponds to a separate Python function or module, making the mapping from diagram to code straightforward.

E. Stage 5: MedProc – Medical Information Processing

Stage 5 processes the medical history and report files to derive Evolution (E) and structured clinical features. The MedProc diagram in Fig. 6 shows an elongated container with two textual inputs feeding into a central NLP pipeline.

The two top **input blocks** correspond to:

- **Input: Medical History (Text)**: free-text description provided by the patient or clinician.
- **Input: Report ≤ 5 MB**: uploaded investigation report (PDF) or referral note.

Both streams converge into the **Preproc** block, which performs OCR (for PDFs), cleanup, and tokenization with an explicit constraint of ≤ 512 tokens to fit BioBERT-style models. The next block, **BioBERT Clinical NLP**, denotes the fine-tuned BioBERT encoder responsible for entity recognition (symptoms, body sites, durations) and relation extraction (e.g., symptom progression).

Finally, the **output block** at the bottom of Fig. 6 produces a JSON structure containing:

- Extracted symptoms and risk factors.
- A suggested high-level textual recommendation at general physician level (non-specialist).
- An Evolution (E) value summarizing how much the lesion has changed over time.

Thus, the diagram clarifies that MedProc is itself a mini-pipeline: OCR and cleanup, transformer-based NLP, and rule-based post-processing, all encapsulated in a single stage that outputs a clean JSON message.

F. Stage 6: IT-Fusion – Final ABCDE and Risk Computation

Stage 6 combines MedProc outputs with HC outputs to compute a final risk estimate. Fig. 7 depicts this stage as a symmetric fusion container.

At the top left, the **Input: MedProc JSON** block provides symptoms and E-value from Stage 5. At the top right, the **Input: HC Output** block supplies the class label and ABCD values from Stage 4. Both feed into the **Preproc: Feature Alignment** block, which converts heterogeneous fields into a joint feature representation:

- Numerical features (ABCD and E) are normalized to a fixed scale.
- Categorical features (class labels, symptom categories) are embedded as one-hot or learned embeddings.
- Textual evidence is tokenized for the multimodal transformer.

The aligned features are then fed into the **MARIA Model** block, indicating a MARIA-style transformer that performs image–text fusion and learns how to weight visual and textual evidence [56]. The final **Output** block yields a *Risk JSON* containing:

- A calibrated risk score (e.g., probability of malignancy).
- The fused ABCDE values.
- An evidence list describing which signals (e.g., high asymmetry, bleeding, strong family history) contributed most to the risk.

From an implementation perspective, the diagram shows that IT-Fusion is not just a single model call: it is explicitly preceded by a feature-alignment step that encodes and concatenates MedProc and HC outputs before passing them into the transformer.

G. Stage 7: Visual Question Answering (VQA)

Stage 7 is responsible for converting the fused evidence into natural-language answers. Fig. 8 expands the VQA pipeline.

The diagram shows three inputs at the top:

- **Input: ROI (224×224)**: the cropped lesion image.
- **Input: Question ≤ 128 tokens**: the current user query from the chat interface.
- **Input: IT JSON**: the risk JSON from Stage 6, containing ABCDE values, risk scores, and textual evidence.

These feed into a **Preproc** block that tokenizes the question, normalizes the ROI image, and converts the IT JSON into a

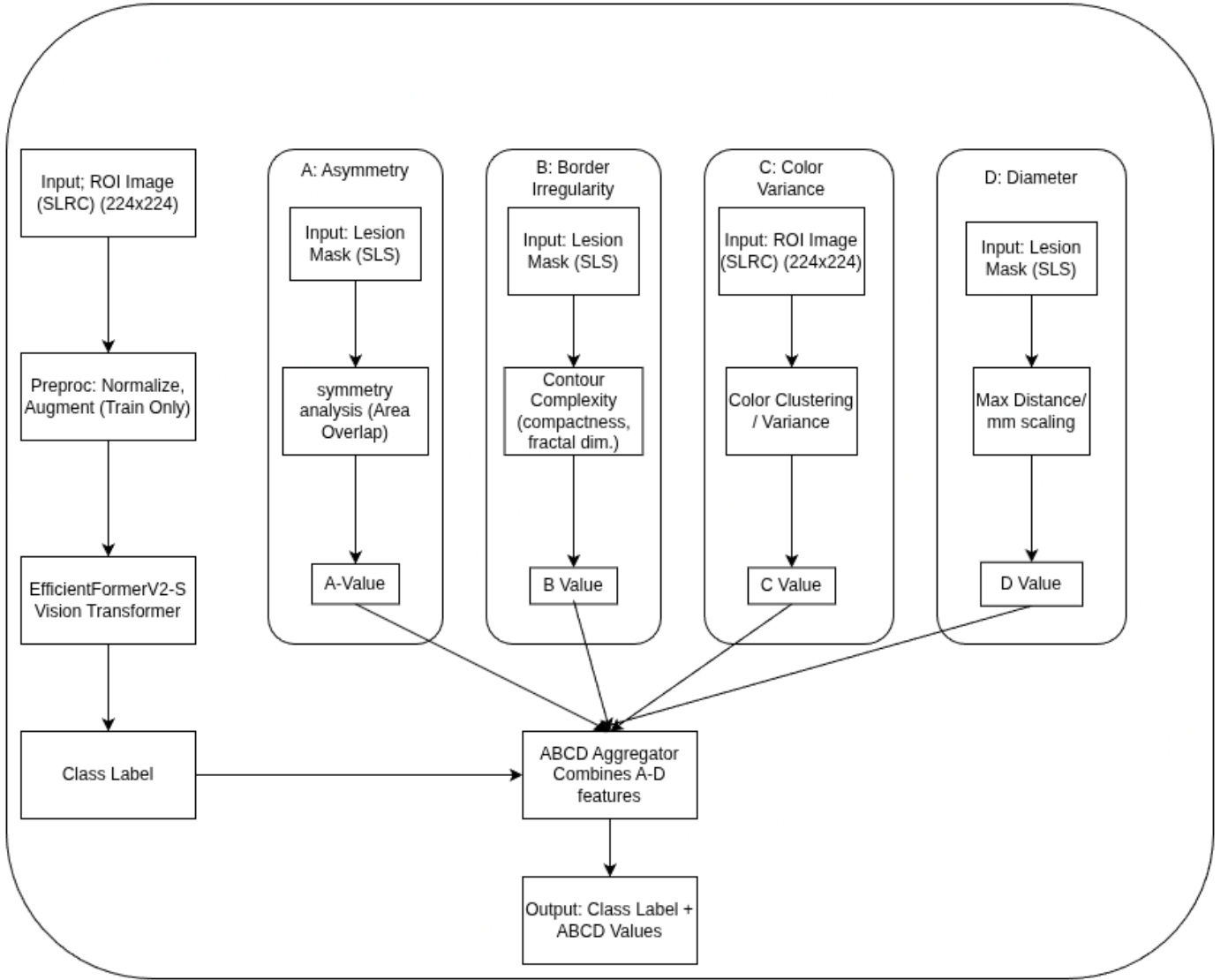


Fig. 5: Stage 4 (HC): EfficientFormerV2-based lesion classification and four parallel sub-pipelines computing A, B, C, and D values from the SLS mask and ROI image.

textual or structured prompt (e.g., “A=2.3, B=1.8, C=3.0, D=6 mm, E=high”).

The core **R-LLaVA** block represents an R-LLaVA-style vision–language model that attends jointly to the image and the text inputs [38]. The bottom **Output** block explicitly states that the stage returns both the final natural-language answer and an *evidence explanation*, which can be surfaced in the UI as bullet points or a short paragraph.

At implementation level, the diagram makes clear that VQA is invoked only after IT-Fusion has produced a consolidated JSON; the model is therefore constrained to answer using grounded, ABCDE-aware evidence rather than hallucinating from the raw image alone.

H. Stage 8: Frontend–Backend and API Layer

Stage 8 exposes the system as a chat-based web application with:

- A React-based frontend for uploading lesion images, attaching reports, and entering questions.
- A FastAPI backend orchestrating SLS, SLRC, HC, Med-Proc, IT-Fusion, and VQA as modular microservices.
- Secure storage and retrieval of intermediate artifacts (e.g., masks, JSON, logs) for auditability.

Design considerations are informed by dermatology chatbot case studies and conversational diagnostic LLM frameworks that highlight UX, disclaimers, and escalation pathways [40], [41], [43], [44].

I. Stage 9: Testing, Monitoring, and Deployment

Stage 9 focuses on:

- Unit and integration tests for all stages.
- Offline evaluation on held-out datasets (see Section V).
- Continuous monitoring for drift, error patterns, and user feedback.

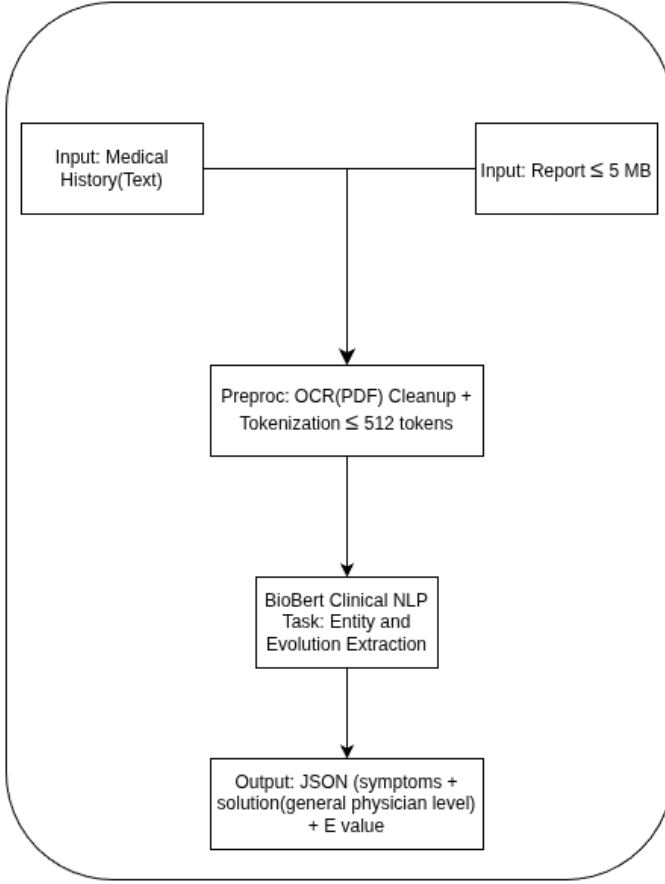


Fig. 6: Stage 5 (MedProc): OCR and cleanup of text inputs followed by BioBERT-based clinical NLP to extract entities, symptoms, and Evolution (E) values.

- Deployment to a controlled environment (e.g., internal research server) with appropriate access controls and logging.

V. DATASETS AND EXPERIMENTAL DESIGN

A. Stage-Wise Dataset Allocation

Table I summarizes the planned datasets per stage, drawing on standard dermatoscopic collections and clinical text corpora [14]–[17], [20], [24], [25], [34]–[36].

B. Data Curation and Preprocessing

Because datasets originate from different centers with varying acquisition protocols, we will:

- Harmonize label taxonomies (e.g., mapping to a common ontology such as DermIM’s hierarchy [20]).
- Standardize image resolutions and color profiles.
- Create clean splits where patient-level leakage is minimized.
- For VQA, generate dermatology-specific question–answer pairs from DermIM captions and literature where necessary, following methods that synthesize QA from medical texts [54].

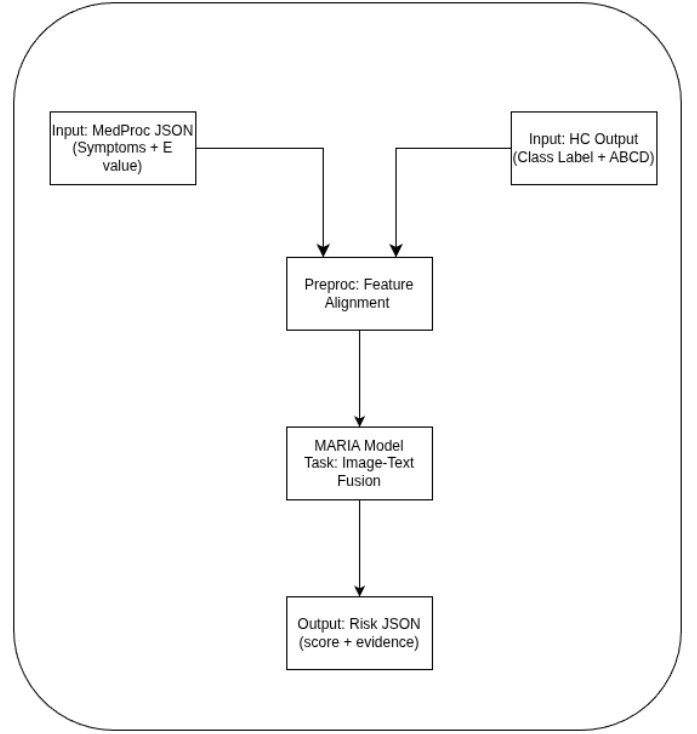


Fig. 7: Stage 6 (IT-Fusion): alignment of MedProc JSON and HC outputs, followed by MARIA-style image–text fusion to produce a risk JSON with score and evidence.

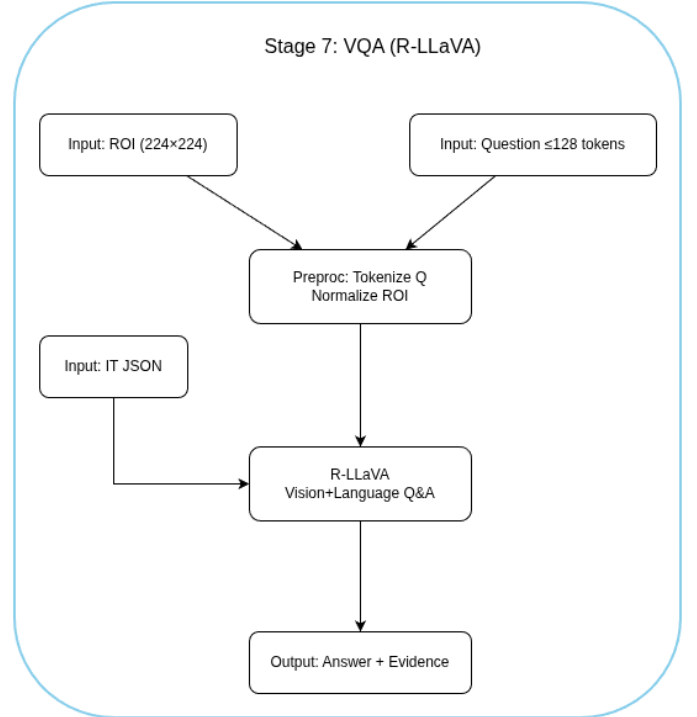


Fig. 8: Stage 7 (VQA): R-LLaVA-based multimodal Q&A over ROI image, user question, and IT-Fusion JSON to generate answers with evidence.

TABLE I: Planned datasets for each stage in NutriDermAI. Counts are approximate and may be refined during implementation.

Stage	Task	Dataset(s)	Train	Val	Test	Total
SLS	Segmentation	ISIC 2016, ISIC 2017 – Segmentation tasks	2,900	400	979	4,279
SLRC	ROI Cropping	ROIs derived from SLS outputs	2,900	400	979	4,279
HC (L1)	Benign vs. malignant	ISIC 2018 classification (HAM10000-style) [15]	7,000	1,015	2,000	10,015
HC (multi)	Extended classification	ISIC 2016–2024, BCN20000, MiLK-10K [14], [16], [17]	$\approx 100,000$ images split 70/10/20			$\approx 100,000$
MedProc	Clinical NLP	MIMIC-III, MIMIC-IV Note [24], [25]	50,000	5,000	10,000	65,000
IT-Fusion	Multimodal	Derm1M + HAM10000 + PAD-UFES-20 + ISIC 2016–2024 + BCN20000 + MIMIC text [14]–[16], [20]	700,000	100,000	200,000	1,000,000
VQA	Med & dermat VQA	DermaVQA, SkinCancerVQA, SkinQA, PathVQA, VQA-RAD, HEALMQA [34]–[37]	80,000	10,000	15,000	105,000

VI. IMPLEMENTATION PLAN

A. Model Choices

For each stage, candidate models are selected based on performance and feasibility with GPUs of around 12 GB VRAM:

- **SLS:** ARC-UNet as primary segmentation backbone, with optional comparison to ScaleFusionNet, ESC-UNet, MRP-UNet, Meta-UNet, BiADATU-Net, MUCM-Net, and CNN-ViT hybrids [7]–[13], [47].
- **HC:** EfficientFormerV2 backbone for hierarchical classification and ABCD-based feature fusion [45], [50].
- **MedProc:** BioBERT-based encoder fine-tuned on MIMIC-III/IV notes, with rule-based ABCDE logic grounded in dermatology literature [1]–[3], [23].
- **IT-Fusion:** MARIA-style multimodal transformer with attention-based modality fusion, extended with rule-based validators [56]–[58].
- **VQA:** An R-LLaVA-style ROI-aware Med-VQA model combined with knowledge-enhanced ensembles and RL-based answer reranking [27], [29], [30], [32], [38], [52], [53], [59], [60].

B. Training Strategy

Each stage is trained separately, followed by joint fine-tuning:

- Pre-train SLS and HC using standard segmentation and classification losses (Dice, cross-entropy, focal loss).
- Train MedProc on clinical NER and relation tasks; validate on manually curated dermatology notes.
- Train IT-Fusion on multimodal samples with ground-truth ABCDE and labels, using calibration-aware losses (e.g., Brier score).
- Train VQA on a mix of general Med-VQA and dermatology-specific VQA datasets, with additional reinforcement signals for answer correctness and format.

Inference will be orchestrated via FastAPI, with GPU-accelerated models containerized (e.g., Docker) and exposed via internal APIs.

VII. EVALUATION PLAN

A. Stage-Specific Metrics

We define metrics per stage:

- **SLS:** Dice coefficient, Intersection-over-Union (IoU), and boundary F1 scores on ISIC segmentation benchmarks [14], [46].
- **SLRC:** Accuracy of lesion coverage (percentage of lesion pixels inside ROI) and preservation of border areas.
- **HC:** Accuracy, balanced accuracy, AUC-ROC, F1-score for benign vs. malignant and multi-class tasks [15]–[17].
- **ABCD/ABCDE:** Correlation and mean absolute error between computed ABCDE values and dermatologist-provided scores where available; agreement on risk bands used in ABCD-based scoring systems [1]–[4].
- **MedProc:** Precision, recall, F1 for clinical entities and attributes (e.g., itching, bleeding, duration).
- **IT-Fusion:** Calibration metrics (Brier score, expected calibration error), risk-curve analyses, and overall AUC-ROC on malignant vs. benign classification with and without text features [21], [22].
- **VQA:** BLEU, ROUGE-L, and answer exact match metrics, plus specialized metrics used in HEALMQA and Med-VQA challenges, alongside human evaluation by dermatologists for correctness and helpfulness [34]–[37], [55].

B. End-to-End Evaluation

End-to-end evaluation includes:

- Measuring improvement in malignant lesion recall when integrating MedProc and IT-Fusion compared to image-only baselines.
- Assessing answer faithfulness by verifying whether VQA rationales are consistent with ABCDE and fusion-stage JSON.
- Conducting small-scale user studies with dermatologists to assess usability, explanation quality, and potential for workflow integration, informed by AMIE and dermatology chatbot studies [40]–[44].

VIII. ETHICAL, SAFETY, AND DEPLOYMENT CONSIDERATIONS

Deploying NutriDermAI in clinical or patient-facing settings requires robust safeguards:

- **Scope and disclaimers:** The system is positioned as decision support rather than a standalone diagnostic tool.
- **Escalation and uncertainty handling:** When confidence is low or high-risk patterns are detected, the system explicitly recommends dermatologist consultation and avoids overconfident statements.

- **Privacy:** Image and text data are processed under appropriate anonymization and secure storage guidelines.
- **Bias and fairness:** Evaluation on diverse datasets and continual monitoring for performance disparities across skin tones and demographic subgroups.

These choices are guided by recent reviews of medical MM-LLMs, ChatGPT in dermatology, and large-scale conversational diagnostic systems [33], [42], [43].

IX. CONCLUSION AND FUTURE WORK

This paper proposes NutriDermAI, a multimodal dermatology assistant that unifies lesion segmentation, ROI cropping, hierarchical classification, ABCDE-aware feature computation, clinical text understanding, multimodal fusion, and interactive VQA within a single explainable pipeline. Building on dermatology foundation models, clinical NLP, and medical VQA literature, NutriDermAI aims to provide clinically meaningful, interpretable, and conversational support for skin lesion assessment.

Future work includes:

- Collecting dermatologist-annotated ABCDE scores and longitudinal image sequences for more robust Evolution modeling.
- Exploring parameter-efficient finetuning of large multimodal models to fit resource-constrained deployments.
- Extending the architecture to additional visual dermatology tasks (e.g., whole-body screening) and other specialties such as radiology and ophthalmology.
- Conducting prospective clinical trials to quantify impact on diagnostic accuracy, workflow efficiency, and patient experience.

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